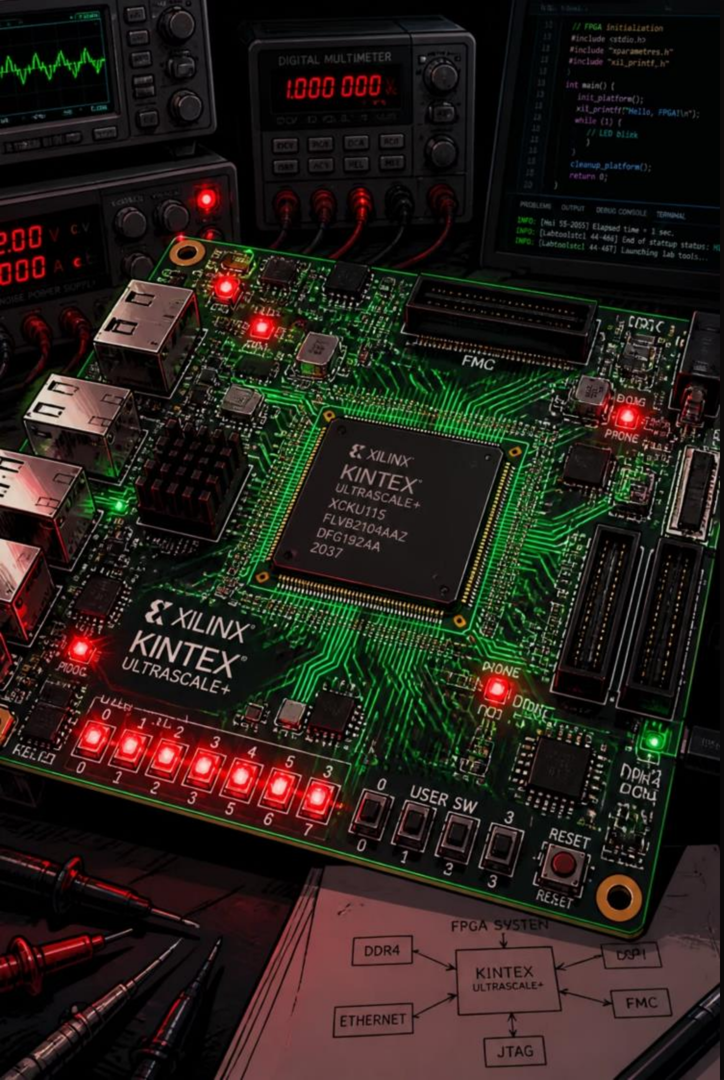

Design and Implementation of a Vending Machine Controller using Verilog

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VLSI DESIGN PROJECT

MOORE FSM ARCHITECTURE

What is a Vending Machine Controller?

A digital system that **accepts coins, verifies payment, allows product selection, dispenses the item, and returns change** all governed by a Moore Finite State Machine implemented in Verilog HDL.

Verilog is a Hardware Description Language (HDL) used to model, simulate, and design digital electronic systems.

Railway & Airports

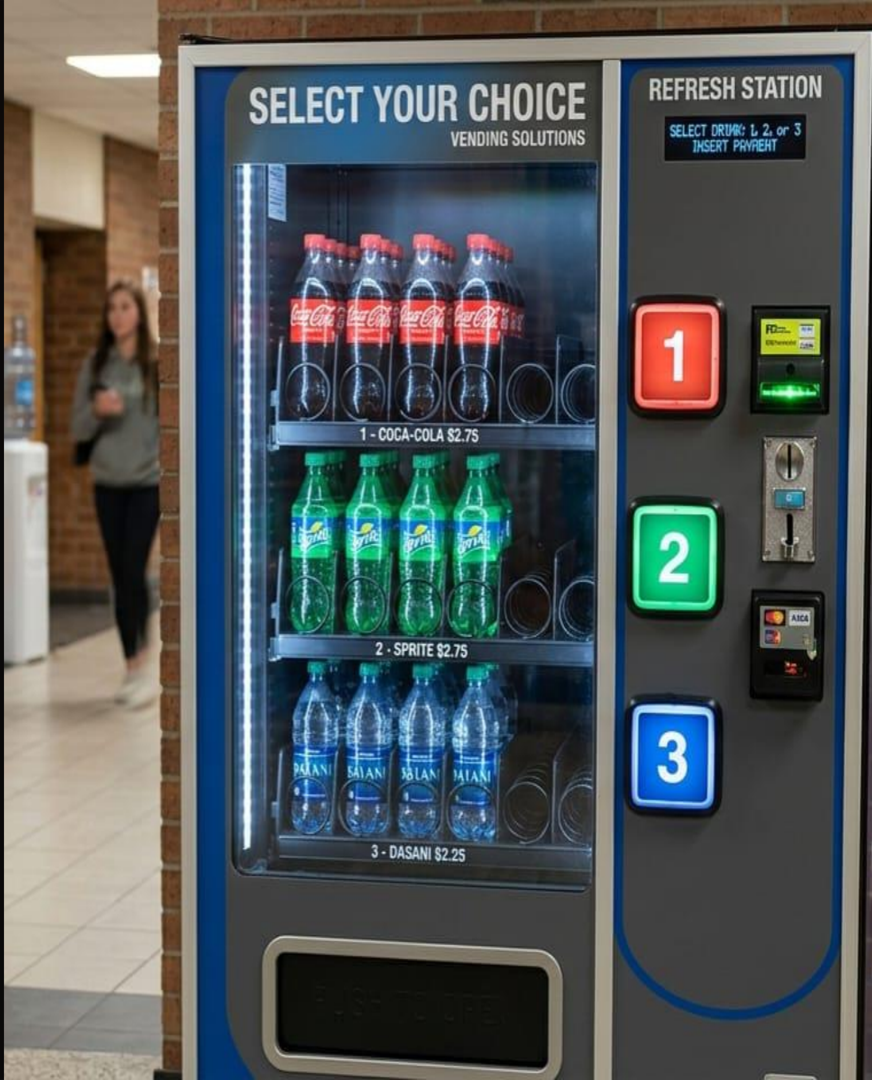
High-traffic automated retail

Schools & Hospitals

Reliable, low-maintenance access

Malls & Offices

24/7 unattended service



Problem Statement & Objectives

The Challenge

- Accurate payment verification under all conditions
- Fast, error-free change calculation
- Minimal human intervention
- Continuous, reliable operation

Our Objective

Design an **FPGA-based controller** that:

- Accepts ₹5, ₹10, and ₹20 coins
- Supports four product selections
- Calculates and returns change automatically
- Displays balance on a seven segment display
- Dispenses the selected product reliably

Tools & Software Used

Design & Simulation

- **Verilog HDL** — RTL source code for FSM, datapath, and display driver
- **Xilinx Vivado** — Synthesis, implementation, and FPGA bitstream generation
- **ModelSim** — Functional simulation of testbenches
- **ISim** — Waveform visualization for debugging signal transitions

Verification & Hardware

- **Testbench** — Self-checking Verilog testbench covering all coin sequences, product selections, and edge cases (reset, under/over payment)
- **FPGA Board** — Physical deployment for I/O validation (switches for coins, LEDs for dispense, 7-segment for balance)
- **Constraints File (.xdc)** — Pin assignments mapping Verilog ports to physical FPGA pins

All modules are written in synthesizable Verilog no system tasks or delays in RTL code.

System Workflow

The controller follows a deterministic sequence governed by a Moore FSM outputs depend only on the current state, ensuring glitch free operation.

Select Product

Insert Coins

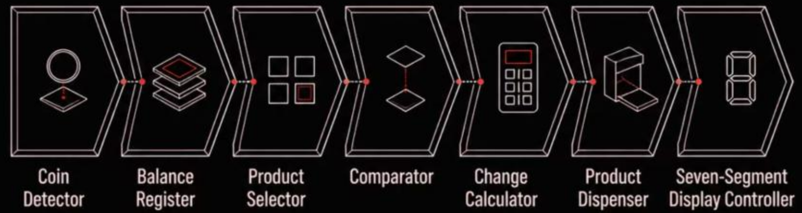
Update Balance

Compare Balance

Dispense Product

Registers store the running balance, the comparator validates payment sufficiency, and the arithmetic unit computes exact change all synchronized to the system clock.

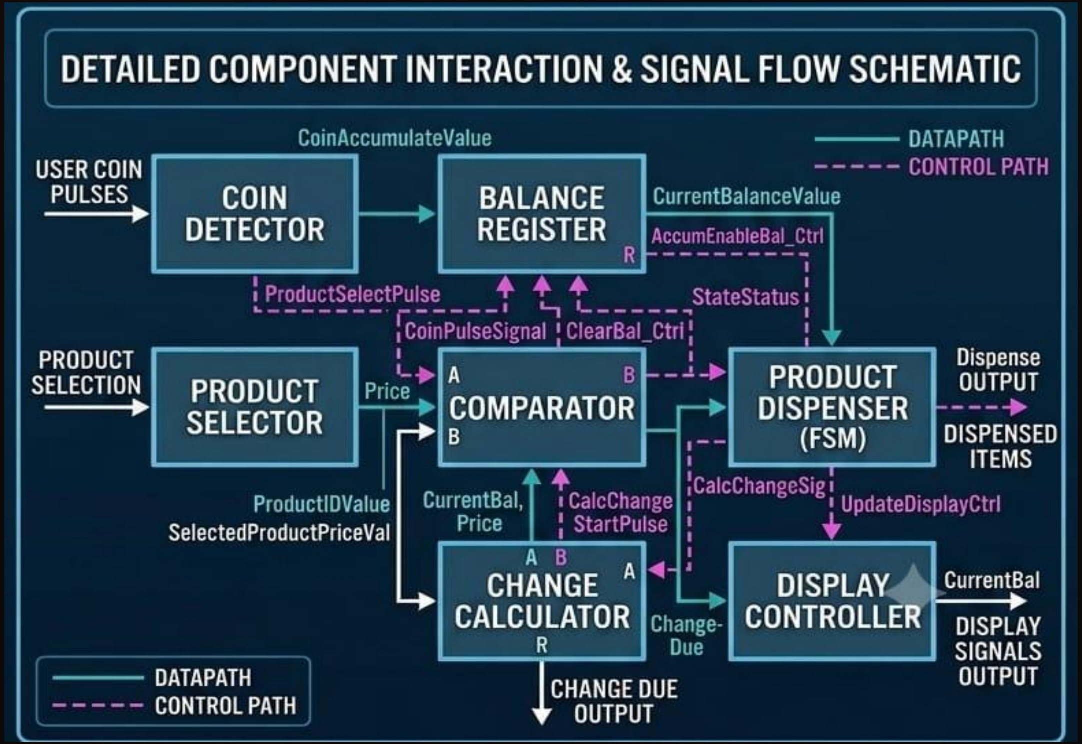
Architectural Block Diagram



Key Modules

- **Coin Detector** — Identifies ₹5, ₹10, ₹20 inputs
- **Balance Register** — Accumulates inserted value
- **Comparator** — Checks balance ≥ product price
- **Change Calculator** — Computes balance - price
- **Display Controller** — Drives seven-segment output

Design Features



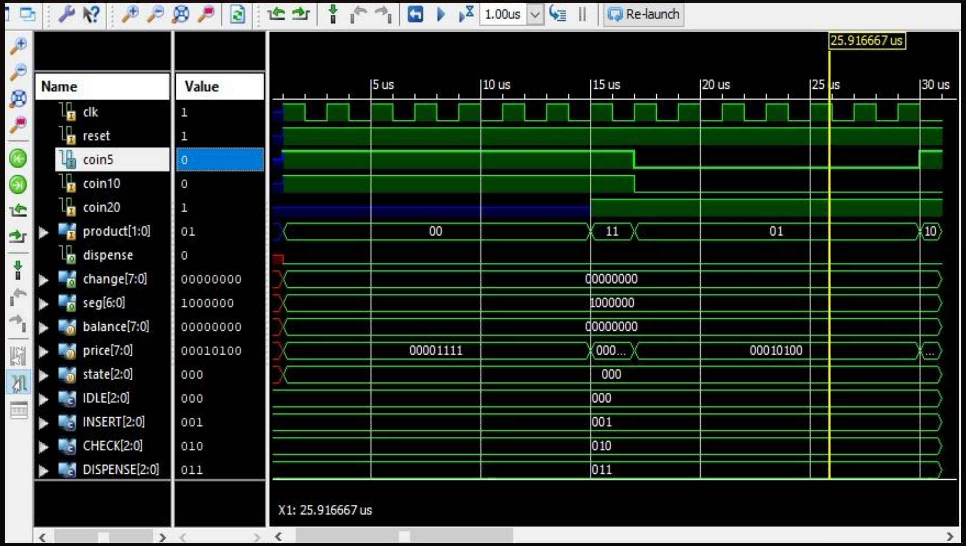
Hardware Architecture

- Moore Finite State Machine — state outputs are stable
- Structural Verilog HDL — modular, synthesizable code
- FPGA-compatible design — portable across platforms

Functional Capabilities

- Three coin denominations: ₹5, ₹10, ₹20
- Four selectable products with distinct prices
- Automatic balance update and change return
- Seven-segment display for real-time balance feedback
- Asynchronous reset for system initialization

Verilog Design & Simulation Results



Implementation Stack

Simulated using **Xilinx Vivado / ISE** all testbenches verified for corner cases including exact payment, overpayment, and insufficient funds.

Verification Checklist

- ✓ Coin detection and denomination recognition
- ✓ Balance register updates correctly
- ✓ Comparator validates payment threshold
- ✓ Product dispense signal asserted
- ✓ Change calculated and displayed accurately

For product reference:
// Product Selection
// 00 = Chips (15)
// 01 = Coke (20)
// 10 = Biscuit (25)
// 11 = Chocolate (30)

Moore FSM

State encoding & transitions

Registers

Balance accumulation

Comparator

Price validation

7-Seg Decoder

BCD to display output

Advantages & Limitations

Advantages

- Fast, glitch-free FSM-driven response
- Accurate arithmetic with zero rounding errors
- Fully automated — no manual intervention
- Modular, reusable Verilog architecture
- Low hardware complexity on FPGA fabric
- Easily expandable for additional features

Current Limitations

- Fixed coin set no custom denomination support
- Limited to four predefined products
- No present design supports sequential currency detection, where only one note or coin is processed at a time.
- No inventory or stock monitoring
- Seven-segment display only no rich UI

Limitations are architectural choices, not design flaws each is addressable in future iterations.

Comparison with Existing Systems

Attribute	Traditional Machine	Proposed Verilog Controller
Implementation	Mechanical / Relay based	Digital FPGA based
Maintenance	Higher moving parts wear	Low solid-state logic
Processing Speed	Slower, latency prone	High speed, clock-synchronous
Flexibility	Fixed wiring, hard to modify	Reprogrammable via firmware
Architecture	Monolithic, hard-wired	Modular, reusable blocks
Control	Semi-manual	Fully automatic FSM

Our design demonstrates that FPGA based digital control offers superior speed, maintainability, and scalability over legacy mechanical approaches.

[redacted]

Conclusion & Future Scope

Conclusion

The Verilog-based vending machine controller was **successfully designed, implemented, and simulated**. All functional requirements coin acceptance, product selection, balance comparison, dispensing, change calculation, and display were verified across all test scenarios.

SIMULATION VERIFIED FPGA READY

Current Enhancements

- UPI / QR code & NFC payment integration
- LCD / TFT display with touchscreen UI
- IoT-based remote monitoring & analytics
- Product inventory management system
- Mobile app connectivity & voice guidance